Neural Networks and Gradient Descent Recap

Gradient descent is a first-order iterative optimization algorithm for finding a local minimum of a differentiable function. To find a local minimum of a function using gradient descent, we take steps proportional to the negative of the gradient (or approximate gradient) of the function at the current point.

In machine learning, we use gradient descent to update the parameters of our model. Parameters refer to coefficients in Linear Regression and weights in neural networks. This relies on the Backpropagation algorithm, which calculates the gradient of the loss function with respect to the weights of the network.

Attention Mechanism in Transformers

The self-attention mechanism allows the inputs to interact with each other and compute a representation of each input word that models the contextual meaning. In the transformer, self-attention computes a set of queries, keys, and values. It maps a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.